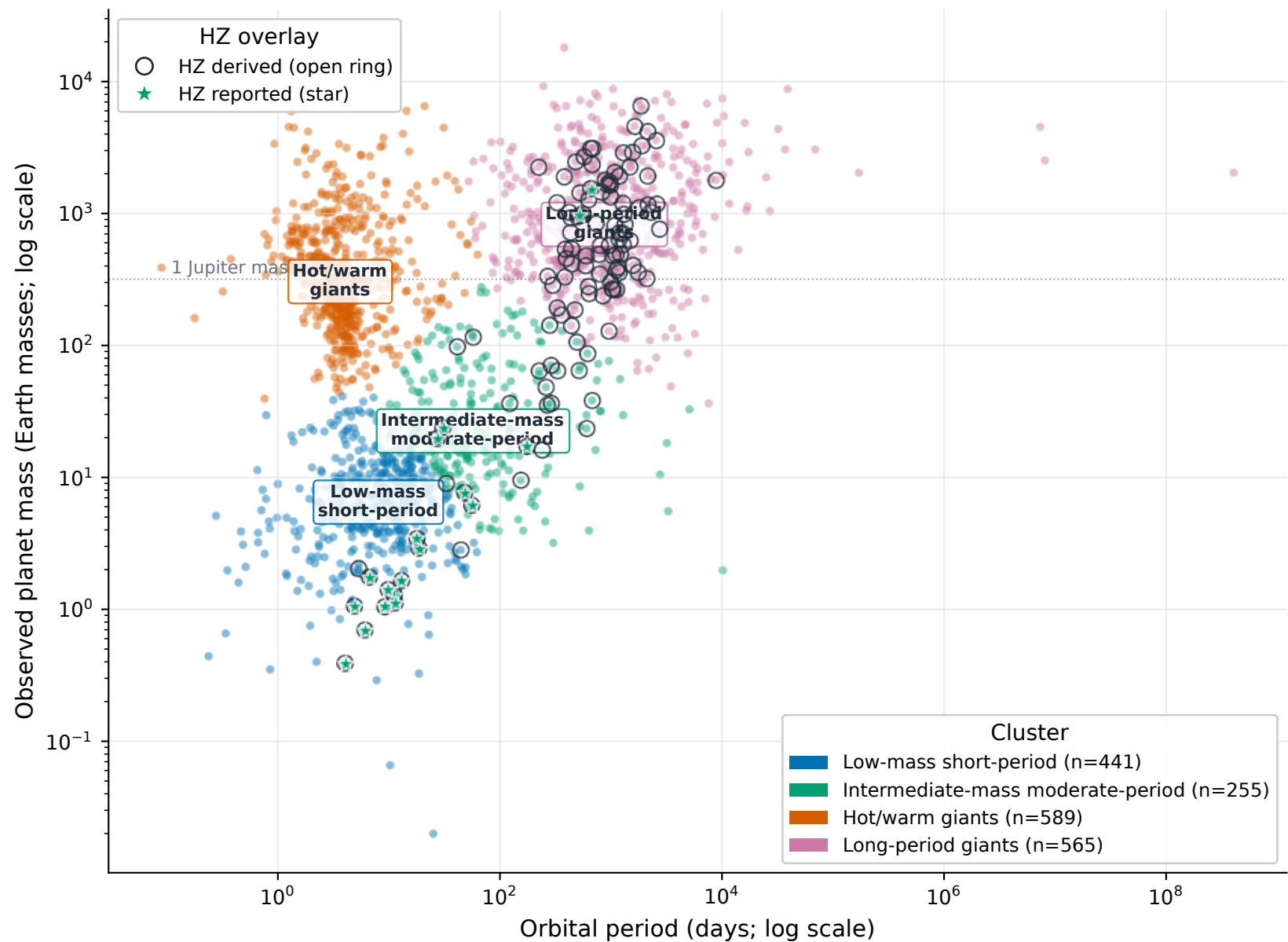


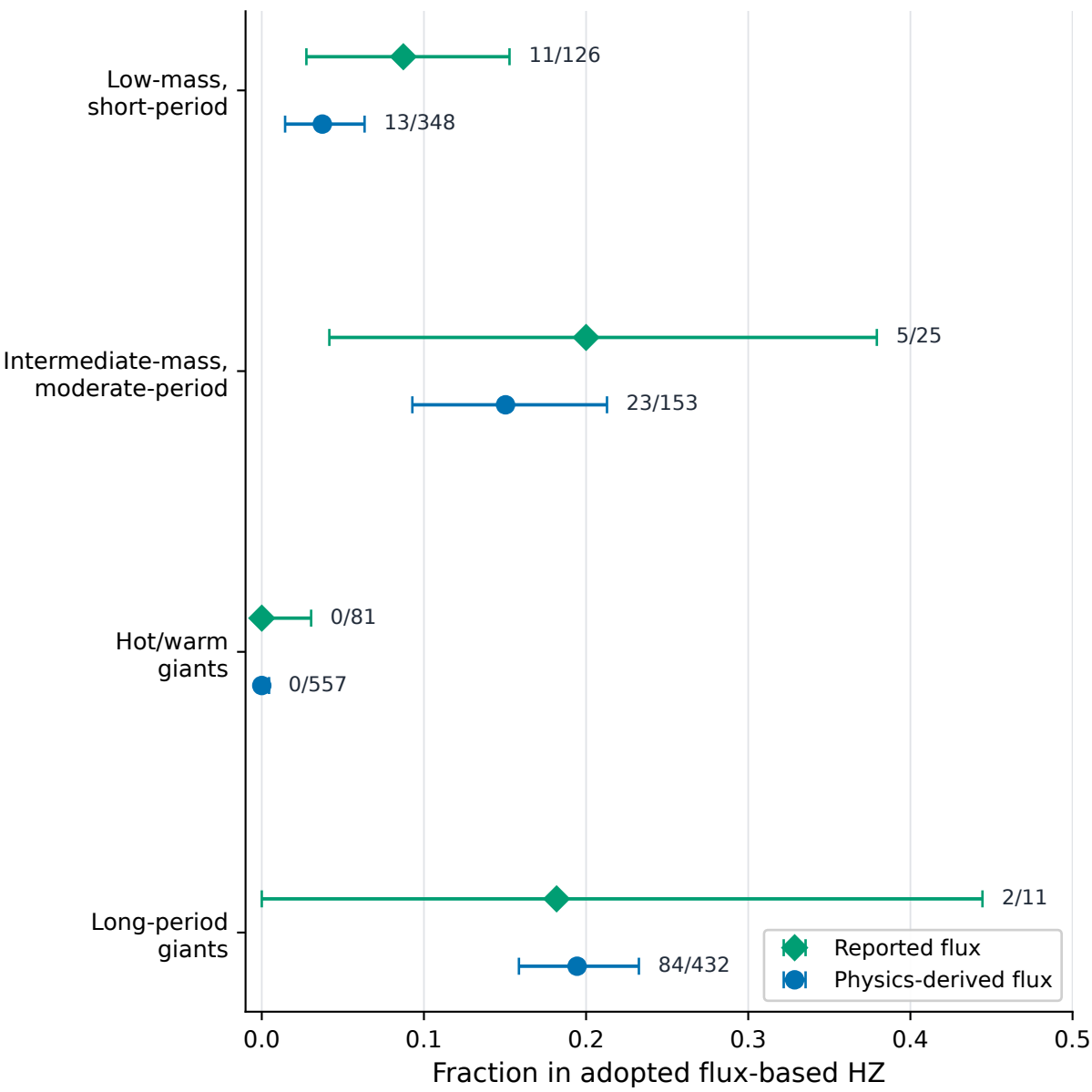
Mass-period clusters separate hot giants from longer-period HZ-overlap groups

K-means used observed planet mass and orbital period only; neither reported nor derived HZ status entered the clustering.

A. Four mass-period archetypes



B. HZ overlap with 95% uncertainty



Power: reported overlap uses 243 planets with 18 HZ positives across only 13 hosts; derived overlap uses 1,490 planets with 120 positives across 113 hosts.

Intervals use host-level percentile bootstrap; zero-event cells use Jeffreys intervals. Cluster names are descriptive, and a flux-based HZ giant is not Earth-like or necessarily surface-habitable.

Planet radius is absent, so super-Earths and sub-Neptunes cannot be cleanly separated in this dataset.